1. Find each of the following limits: (1) $\lim\limits_{(x,y)\to(0,1)}\dfrac{1-xy}{x^2+y^2}$ (2) $\lim\limits_{(x,y)\to(0,0)}\dfrac{2-\sqrt{xy+4}}{xy}$ (3) $\lim\limits_{(x,y)\to(0,0)}\dfrac{1-\cos(x^2+y^2)}{(x^2+y^2)e^{x^2y^2}}$

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2. Find the partial derivatives of the following functions: (1) $z=\ln\left(\tan\dfrac{x}{y}\right)$ (2) $u=x^{\frac{y}{z}}$
3. Find $\dfrac{\partial^2 z}{\partial x^2}$, $\dfrac{\partial^2 z}{\partial y^2}$ and $\dfrac{\partial^2 z}{\partial x\partial y}$ for the function $z=y^x$
4. Let $f(x,y,z)=xy^2+yz^2+zx^2$; find $f_{xx}(0,0,1)$, $f_{xx}(1,0,2)$, $f_{yz}(0,-1,0)$ and $f_{zzx}(2,0,1)$.

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5. Find the total differentials of the following functions: (1) $z=\sin(x\cos y)$ (2) $z=e^{\frac{y}{x}}$
6. Find the total differential of the following function at each of the given points: $z=\dfrac{x}{\sqrt{x^2+y^2}}$ at the points $(1,0)$ and $(0,1)$

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7. Let $z=e^{x-2y}$, where $x=\sin t$ and $y=t^3$; find $\dfrac{dz}{dt}$.
8. Let $z=(x^2+y^2)^{xy}$; find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$.
9. Find the first-order partial derivatives of the following function (where $f$ has continuous first-order partial derivatives): $u=f(x^2-y^2,xy)$
10. Let $u=f(x+y+z,x^2+y^2+z^2)$, where $f$ has continuous second-order partial derivatives; find $\Delta u=\dfrac{\partial^2 u}{\partial x^2}+\dfrac{\partial^2 u}{\partial y^2}+\dfrac{\partial^2 u}{\partial z^2}$.

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11. Let $\sin y+e^x-xy^2=0$; find $\dfrac{dy}{dx}$.
12. Let $\dfrac{x}{z}=\ln\dfrac{z}{y}$; find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$.
13. Let $e^z-xyz=0$; find $\dfrac{\partial^2 z}{\partial x^2}$.
14. Find the derivatives or partial derivatives of the functions determined by the following systems of equations: (1) Let $\begin{cases}z=x^2+y^2\\x^2+2y^2+3z^2=20\end{cases}$; find $\dfrac{dy}{dx}$ and $\dfrac{dz}{dx}$. (3) Let $\begin{cases}u=f(ux,v+y)\\v=g(u-x,v^2y)\end{cases}$, where $f,g$ have continuous first-order partial derivatives; find $\dfrac{\partial u}{\partial x}$ and $\dfrac{\partial v}{\partial x}$.

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15. Find the tangent line and normal plane of the curve $\boldsymbol{r}=\boldsymbol{f}(t)=(t-\sin t)\boldsymbol{i}+(1-\cos t)\boldsymbol{j}+\left(4\sin\dfrac{t}{2}\right)\boldsymbol{k}$ at the point corresponding to $t_0=\dfrac{\pi}{2}$.
16. Find the tangent line and normal plane of the curve $\begin{cases}x^2+y^2+z^2-3x=0\\2x-3y+5z-4=0\end{cases}$ at the point $(1,1,1)$.

17. Find the tangent planes of the ellipsoid $x^2 + 2y^2 + z^2 = 1$ that are parallel to the plane $x - y + 2z = 0$ .

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18. Let $a, b$ be real numbers. Among the directional derivatives of the function $z = 2 + ax^2 + by^2$ at the point $(3, 4)$, the one along the direction $(-3, -4)$ is the largest, with maximum value $10$. Find $a$ and $b$.

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19. Find the extrema of the following functions: (1) $z = x^3 - 4x^2 + 2xy - y^2$ (2) $z = (x^2 + y^2)^2 - 2x^2 + 2y^2$